

AE_544_LecNote05_Lagrangian_Dynamics_Ch05

Disclaimers

This repository contains markdown-based lecture notes for the Spring 2025 semester. These materials are intended solely for students enrolled in the course during this term. Mistakes may exist and will be corrected as they are identified, but the official textbook remains the ultimate reference for accuracy.

Generalized Coordinates

Even in this simple and most familiar example, we see that **an infinity of coordinate choices** is possible. Depending on the **objectives being pursued** in any given problem, any of these coordinate choices may be **appropriate**. It is clear that the details of most traditional analyses, such as formulating the differential equations of motion, are affected by the coordinates selected, because expressions for all kinematical and physical quantities depend on the coordinate choice.

Develop **a universal form of the differential equations of motion**, as a function of the system kinetic energy and unspecified generalized coordinates, i.e., $T(q_1, q_2, \dots, q_n; \dot{q}_1, \dot{q}_2, \dots, \dot{q}_n)$, that holds for all infinity of possible coordinate choices, and for particle motions, rigid body motions, translations, rotations, deformational vibrations, etc.

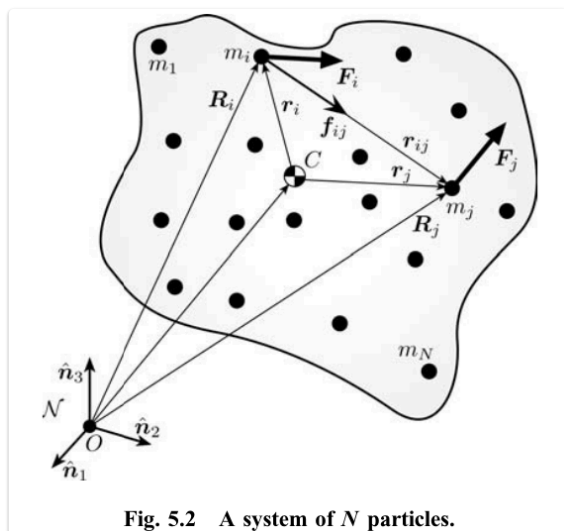
At the heart of these developments, it is evident that the various vector descriptions for position $\mathbf{r}$, velocity $\dot{\mathbf{r}}$, and acceleration $\ddot{\mathbf{r}}$ are alternative descriptions or mathematical representations of **the same physical quantities**.

Although broad generality necessarily **introduces a level of abstraction in the formulation**, the ensuing analysis is of bearable complexity and well justified by the powerful generalized results obtained therefrom.

D'Alembert's Principle (a reformulation of Newton's second law)

✓ The most important role of d'Alembert's principle is that it is a stepping stone leading to Lagrange's equations, Hamilton's principle, and other variational principles in analytical dynamics.

Virtual Displacements and Virtual Work



For a system of N particles, the total force vector acting on m_i to be segregated into two summed subsets of forces as

$$\mathbf{F}_i = \mathbf{f}_{c_i} + \mathbf{f}_i + \sum_{j=1, i \neq j}^N \mathbf{f}_{ij} \quad (5.8)$$

where

- $\mathbf{f}_{c_i}$ is the vector sum of all virtually nonworking constraint forces (as explained in the following).
- $\mathbf{f}_{ij}$ is the system internal force acting on the i -th due to the j -th particle.

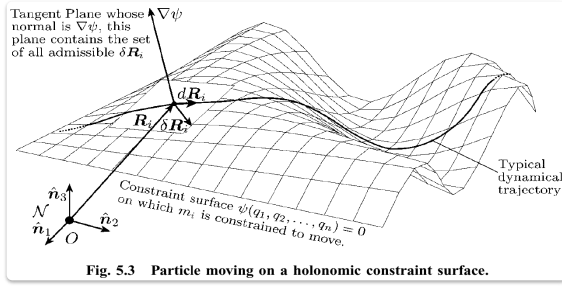
- $\mathbf{f}_i$ is the vector sum of all other forces except the total constraint force $\mathbf{f}_{c_i}$.

We will show that the constraint forces $\mathbf{f}_c$ can be eliminated from the analysis, and this is an advantageous feature common to all of the methods of generalized mechanics.

To accomplish the elimination of the constraint forces, we introduce the concept of virtual displacement $\delta\mathbf{R}_i$. A virtual displacement, in the most general context, is an instantaneous differential displacement for the sake of analysis.

📌 The virtual displacement $\delta(\cdot)$ of a dynamical motion variable $(\cdot)$ is closely related to the first variation of coordinates in variational calculus, which we will discuss in the later semester.

If the constraints acting on the system are smooth differential functions of $\mathbf{R}_i(t, q_1, q_2, \dots, q_n)$, then admissible $\delta\mathbf{R}_i$ are constrained to lie on a smooth holonomic (function of position coordinates only) constraint surface $\psi(t, q_1, q_2, \dots, q_n)$, and we see that admissible or consistent virtual displacements $\delta\mathbf{R}_i$ locate points in a tangent plane, whose normal can be obtained by taking the gradient $\nabla\psi$ of the constraint surface.



Note that the differential displacement $d\mathbf{R}_i$ (total derivative) is tangent to a particular trajectory, whereas the consistent virtual displacement $\delta\mathbf{R}_i$ is an arbitrary differential displacement to any neighboring point in the tangent plane of feasible displacements.

Thus the virtual displacements are not necessarily tangent to any solution trajectory, but they are required to locate neighboring differentially displaced points satisfying the constraints, at some arbitrary and unspecified time t in the motion.

The constraint force $\mathbf{f}_{c_i}$ is always normal to the constraint surface (i.e., in the direction of $\nabla\psi$), and therefore can be written as

$$\mathbf{f}_{c_i} = \lambda \nabla\psi \quad (5.9)$$

where λ is referred to as a Lagrange multiplier.

The **Virtual Work** δW is an abstract idea analogous to mechanical work, but associated with the instantaneous virtual displacements. The virtual work δW done on m_i as a consequence of virtual displacement $\delta\mathbf{R}_i$ is defined as

$$\delta W_i \equiv \mathbf{F}_i \cdot \delta\mathbf{R}_i \quad (5.10)$$

Observe that the constraint force $\mathbf{f}_{c_i}$ is normal to the plane containing all infinity of admissible virtual displacements $\delta\mathbf{R}_i$, and this can be stated as the orthogonality condition:

$$\delta W_{c_i} = \mathbf{f}_{c_i} \cdot \delta\mathbf{R}_i = 0 \quad (5.11)$$

🔧 **Virtual Work** free up the concept of work to more abstract and general cases.

We define the total virtual work δW to be the sum of the δW_i , so that

$$\begin{aligned}
\delta W &= \sum_{i=1}^N \mathbf{F}_i \cdot \delta \mathbf{R}_i \\
&= \sum_{i=1}^N \left(\mathbf{f}_{c_i} + \mathbf{f}_i + \sum_{k=1, k \neq i}^N \mathbf{f}_{ik} \right) \cdot \delta \mathbf{R}_i \\
&= \sum_{i=1}^N (\mathbf{f}_{c_i} \cdot \delta \mathbf{R}_i) + \sum_{i=1}^N (\mathbf{f}_i \cdot \delta \mathbf{R}_i) + \sum_{i=1}^N \left(\sum_{k=1, k \neq i}^N \mathbf{f}_{ik} \cdot \delta \mathbf{R}_i \right) \\
&= \sum_{i=1}^N (\mathbf{f}_i \cdot \delta \mathbf{R}_i) + \sum_{i=1}^N \left(\sum_{k=1}^{i-1} \mathbf{f}_{ik} \cdot \delta \mathbf{R}_i \right) + \sum_{i=1}^N \left(\sum_{k=i+1}^N \mathbf{f}_{ik} \cdot \delta \mathbf{R}_i \right) \\
&= \sum_{i=1}^N (\mathbf{f}_i \cdot \delta \mathbf{R}_i) + \sum_{k=1}^{N-1} \left(\sum_{i=k+1}^N -\mathbf{f}_{ki} \cdot \delta \mathbf{R}_i \right) + \sum_{i=1}^{N-1} \left(\sum_{k=i+1}^N \mathbf{f}_{ik} \cdot \delta \mathbf{R}_i \right) \\
&= \sum_{i=1}^N \mathbf{f}_i \cdot \delta \mathbf{R}_i + \sum_{i=1}^{N-1} \sum_{k=i+1}^N \mathbf{f}_{ik} \cdot (\delta \mathbf{R}_i - \delta \mathbf{R}_k)
\end{aligned} \tag{5.13}$$

Assuming the rigidly constraint (rigid-body assumption) so that $\mathbf{f}_{ik}$ is always normal to $(\delta \mathbf{R}_i - \delta \mathbf{R}_k)$, then we can simplify it to

$$\delta W = \sum_{i=1}^N \mathbf{f}_i \cdot \delta \mathbf{R}_i \tag{5.17}$$

So, both the **nonworking constraint forces** $\mathbf{f}_{c_i}$ as well as the **nonworking system internal forces** $\mathbf{f}_{ik}$ do not contribute to the total virtual work calculation.

① The rigid-body assumption doesn't require $\delta \mathbf{R}_i - \delta \mathbf{R}_k = 0$, but just requires

$$(\mathbf{R}_i - \mathbf{R}_k) \cdot (\delta \mathbf{R}_i - \delta \mathbf{R}_k) = 0 \tag{5.15}$$

meaning that the relative virtual displacement are orthogonal to the relative position vector, and thus the internal force $\mathbf{f}_{ik}$. (Refer to textbook for details.)

Development of d'Alembert's Principle using Virtual Work

$$\mathbf{F}_i = \mathbf{f}_{c_i} + \mathbf{f}_i + \sum_{j=1, j \neq i}^N \mathbf{f}_{ij} \tag{5.8 copied}$$

Applying Newton's second law for the motion of m_i , we have

$$\mathbf{f}_{c_i} + \mathbf{f}_i + \sum_{j=1, j \neq i}^N \mathbf{f}_{ij} - m_i \ddot{\mathbf{R}}_i = 0 \tag{5.18}$$

Upon taking the dot product of Eq. (5.18) with an arbitrary consistent/admissible virtual displacement $\delta \mathbf{R}_i$ and summing over all N particles, we find the most general form of d'Alembert's principle to be:

$$\delta W - \sum_{i=1}^N m_i \ddot{\mathbf{R}}_i \cdot \delta \mathbf{R}_i = 0 \tag{5.19}$$

Consider $\delta \mathbf{R}_i$ to be generated by a set of independent virtual variations in the q_i through

$$\delta \mathbf{R}_i = \sum_{j=1}^n \frac{\partial \mathbf{R}_i}{\partial q_j} \delta q_j \tag{5.20}$$

✍ The number of particles m_i is N . The number of general variables q_j is n .

$$\delta W = \sum_{i=1}^N \mathbf{f}_i \cdot \delta \mathbf{R}_i + \sum_{i=1}^{N-1} \sum_{k=i+1}^N \mathbf{f}_{ik} \cdot (\delta \mathbf{R}_i - \delta \mathbf{R}_k) \tag{5.13 copied}$$

As a consequence, the virtual work δW in Eq. (5.17) can be rewritten as

$$\begin{aligned}
\delta W &= \sum_{i=1}^N \mathbf{f}_i \cdot \delta \mathbf{R}_i \\
&= \sum_{i=1}^N \mathbf{f}_i \cdot \left(\sum_{j=1}^n \frac{\partial \mathbf{R}_i}{\partial q_j} \delta q_j \right) \\
&= \sum_{j=1}^n \left[\sum_{i=1}^N \mathbf{f}_i \cdot \left(\frac{\partial \mathbf{R}_i}{\partial q_j} \delta q_j \right) \right] && \text{(finite series; linearity)} \\
&= \sum_{j=1}^n \left[\sum_{i=1}^N \left(\mathbf{f}_i \cdot \frac{\partial \mathbf{R}_i}{\partial q_j} \right) \delta q_j \right] && \text{(associativity)} \\
&= \sum_{j=1}^n Q_j \delta q_j && (5.21)
\end{aligned}$$

Here, we defined **generalized force** Q_j as a combination of N virtually working forces $\mathbf{f}_i$, which is

$$Q_j \equiv \sum_{i=1}^N \left(\mathbf{f}_i \cdot \frac{\partial \mathbf{R}_i}{\partial q_j} \right) \quad (5.22)$$

Plugging everything back to Eq. (5.19) gives us

$$\sum_{j=1}^n \left[Q_j - \sum_{i=1}^N m_i \ddot{\mathbf{R}}_i \cdot \frac{\partial \mathbf{R}_i}{\partial q_j} \right] \delta q_j = 0 \quad (5.23)$$

Because the δq_j are independent virtual variations, they may be chosen independently and arbitrarily, so each coefficient must be 0 by itself. This leads to the most famous form of **d'Alembert's principle**:

$$\sum_{i=1}^N m_i \ddot{\mathbf{R}}_i \cdot \frac{\partial \mathbf{R}_i}{\partial q_j} = Q_j \quad \text{for } j = 1, 2, \dots, n \quad (5.24)$$

These equations are generally a coupled system of n second-order differential equations.

Observe that the position vector $\mathbf{R}_i(t, q_1, q_2, \dots, q_n)$ can be differentiated to obtain the expression for velocity $\mathbf{V}_i$ as

$$\mathbf{V}_i(t, q_1, q_2, \dots, q_n, \dot{q}_1, \dot{q}_2, \dots, \dot{q}_n) = \dot{\mathbf{R}}_i(t, q_1, q_2, \dots, q_n) = \frac{\partial \mathbf{R}_i}{\partial t} + \sum_{k=1}^n \frac{\partial \mathbf{R}_i}{\partial q_k} \dot{q}_k \quad i = 1, 2, \dots, N \quad (5.25)$$

From this, we can get the following identity (cancellation of dots identity)

$$\frac{\partial \mathbf{V}_i}{\partial \dot{q}_k} = \frac{\partial \dot{\mathbf{R}}_i}{\partial \dot{q}_k} = \frac{\partial \mathbf{R}_i}{\partial q_k} \quad (5.26)$$

Proof: Simply plugging in Eq. (5.25) and solve the partial derivatives. It relies on the fact that $\mathbf{R}_i(t, q_1, q_2, \dots, q_n)$ is not explicitly a function of any $\dot{q}_k$.

We define the **partial velocity** $\mathbf{v}_{ik} \equiv \frac{\partial \dot{\mathbf{R}}_i}{\partial \dot{q}_k} = \frac{\partial \mathbf{R}_i}{\partial q_k}$, so Eq. (5.25) becomes,

$$\mathbf{V}_i = \dot{\mathbf{R}}_i = \frac{\partial \mathbf{R}_i}{\partial t} + \sum_{k=1}^n \dot{q}_k \mathbf{v}_{ik}, \quad i = 1, 2, \dots, N \quad (5.27)$$

The n vectors $\{\mathbf{v}_{i1}, \mathbf{v}_{i2}, \dots, \mathbf{v}_{in}\}$ form a vector basis for the inertial velocity $\mathbf{V}_i$ of the i -th mass m_i , and for the case that time does not appear explicitly, the $\dot{q}_k$ are the coefficients that linearly combine the basis vectors $\mathbf{v}_{ik}$ to give the velocity vector $\mathbf{V}_i$.

The d'Alembert's principle in Eq. (5.24) can be rewritten using generalized forces Q_j as

$$\sum_{i=1}^N m_i \ddot{\mathbf{R}}_i \cdot \mathbf{v}_{ij} = Q_j \quad \text{for } j = 1, 2, \dots, n \quad (5.28)$$

Combine Eqs. (5.22) and (5.29), we have another form of the d'Alembert's principle:

$$\sum_{i=1}^N [\mathbf{f}_i - m_i \ddot{\mathbf{R}}_i] \cdot \mathbf{v}_{ij} = 0 \quad \text{for } j = 1, 2, \dots, n \quad (5.30)$$

where

- $\mathbf{f}_i$ is the vector sum of all other forces acting on m_i

- $\mathbf{V}_i$ is the velocity vector of m_i
- $\mathbf{v}_{ij} = \frac{\partial \mathbf{R}_i}{\partial \dot{q}_j} = \frac{\partial \mathbf{R}_i}{\partial \dot{q}_k}$ is the (Kane's definition) partial velocity (not a velocity decomposition!)

✓ Different forms of d'Alembert's principle:

1. (classical form, Eq. 5.24) $\sum_{i=1}^N m_i \ddot{\mathbf{R}}_i \cdot \frac{\partial \mathbf{R}_i}{\partial \dot{q}_j} = Q_j$
2. $\sum_{i=1}^N m_i \ddot{\mathbf{R}}_i \cdot \mathbf{v}_{ij} = Q_j$
3. (virtual power form, Eq. 5.30) $\sum_{i=1}^N [\mathbf{f}_i - m_i \dot{\mathbf{V}}_i] \cdot \mathbf{v}_{ij} = 0$

where $Q_j = \sum_{i=1}^N \mathbf{f}_i \cdot \frac{\partial \mathbf{R}_i}{\partial \dot{q}_j} = \sum_{i=1}^N \mathbf{f}_i \cdot \mathbf{v}_{ij}$ and $\mathbf{v}_{ij} = \frac{\partial \mathbf{R}_i}{\partial \dot{q}_j} = \frac{\partial \mathbf{R}_i}{\partial \dot{q}_k}$.

Example 5.3: Cart + Pendulum + Spring (Newtonian vs. d'Alembert's principle)

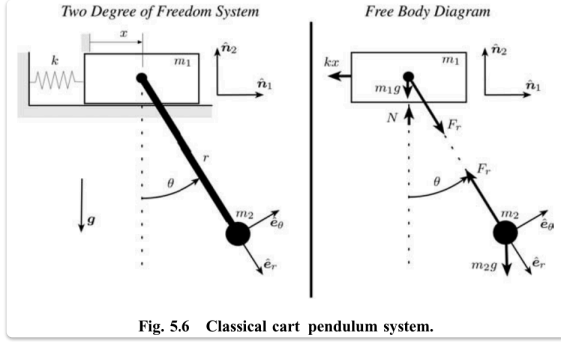


Fig. 5.6 Classical cart pendulum system.

The free body diagram is only needed for direct application of Newton's law.

Kinematics of the two bodies are:

$$\mathbf{R}_1 = x \hat{\mathbf{n}}_1 \quad \dot{\mathbf{R}}_1 = \dot{x} \hat{\mathbf{n}}_1 \quad \ddot{\mathbf{R}}_1 = \ddot{x} \hat{\mathbf{n}}_1 \quad (5.31)$$

$$\begin{aligned} \mathbf{R}_2 &= x \hat{\mathbf{n}}_1 + r \hat{\mathbf{e}}_r = (x + r \sin \theta) \hat{\mathbf{n}}_1 + (-r \cos \theta) \hat{\mathbf{n}}_2 \\ \dot{\mathbf{R}}_2 &= \dot{x} \hat{\mathbf{n}}_1 + r \dot{\theta} \hat{\mathbf{e}}_\theta = (\dot{x} + r \dot{\theta} \cos \theta) \hat{\mathbf{n}}_1 + (r \dot{\theta} \sin \theta) \hat{\mathbf{n}}_2 \\ \ddot{\mathbf{R}}_2 &= \ddot{x} \hat{\mathbf{n}}_1 - r \dot{\theta}^2 \hat{\mathbf{e}}_r + r \ddot{\theta} \hat{\mathbf{e}}_\theta \\ &= (\ddot{x} - r \dot{\theta}^2 \sin \theta + r \ddot{\theta} \cos \theta) \hat{\mathbf{n}}_1 + (r \dot{\theta}^2 \cos \theta + r \ddot{\theta} \sin \theta) \hat{\mathbf{n}}_2 \\ &= (\ddot{x} \sin \theta - r \dot{\theta}^2) \hat{\mathbf{e}}_r + (\ddot{x} \cos \theta + r \ddot{\theta}) \hat{\mathbf{e}}_\theta \end{aligned} \quad (5.32)$$

External forces on the two bodies m_1 and m_2 are

$$\mathbf{f}_1 = -kx \hat{\mathbf{n}}_1 - m_1 g \hat{\mathbf{n}}_2 \quad \mathbf{f}_2 = -m_2 g \hat{\mathbf{n}}_2 \quad (5.35)$$

Differential equations derived via Newton's laws. We need to first introduce the constraint forces N , F_r and then eliminate them. (Refer to textbook page 219).

Differential equations derived via d'Alembert's principle. The chosen generalized coordinates are (x, θ) .

The partial derivatives are

$$\begin{aligned} \frac{\partial \mathbf{R}_1}{\partial x} &= \hat{\mathbf{n}}_1 & \frac{\partial \mathbf{R}_1}{\partial \theta} &= \mathbf{0} \\ \frac{\partial \mathbf{R}_2}{\partial x} &= \hat{\mathbf{n}}_1 & \frac{\partial \mathbf{R}_2}{\partial \theta} &= r \cos \theta \hat{\mathbf{n}}_1 + r \sin \theta \hat{\mathbf{n}}_2 \end{aligned} \quad (5.41)$$

$$Q_j \equiv \sum_{i=1}^N \left(\mathbf{f}_i \cdot \frac{\partial \mathbf{R}_i}{\partial \dot{q}_j} \right) \quad (5.22)$$

Using the definition in Eq. (5.22), we can obtain the generalized forces as

$$\begin{aligned} Q_x &= \mathbf{f}_1 \cdot \frac{\partial \mathbf{R}_1}{\partial x} + \mathbf{f}_2 \cdot \frac{\partial \mathbf{R}_2}{\partial x} \\ &= (-kx \hat{\mathbf{n}}_1 - m_1 g \hat{\mathbf{n}}_2) \cdot \hat{\mathbf{n}}_1 + (-m_2 g \hat{\mathbf{n}}_2) \cdot \hat{\mathbf{n}}_1 = -kx \\ Q_\theta &= \mathbf{f}_1 \cdot \frac{\partial \mathbf{R}_1}{\partial \theta} + \mathbf{f}_2 \cdot \frac{\partial \mathbf{R}_2}{\partial \theta} \\ &= (-kx \hat{\mathbf{n}}_1 - m_1 g \hat{\mathbf{n}}_2) \cdot \mathbf{0} + (-m_2 g \hat{\mathbf{n}}_2) \cdot (r \cos \theta \hat{\mathbf{n}}_1 + r \sin \theta \hat{\mathbf{n}}_2) \\ &= -m_2 g r \sin \theta \end{aligned} \quad (5.42)$$

✓ Different forms of d'Alembert's principle:

1. (classical form, Eq. 5.24) $\sum_{i=1}^N m_i \ddot{\mathbf{R}}_i \cdot \frac{\partial \mathbf{R}_i}{\partial \mathbf{q}_j} = Q_j$
2. $\sum_{i=1}^N m_i \ddot{\mathbf{R}}_i \cdot \mathbf{v}_{ij} = Q_j$
3. (virtual power form, Eq. 5.30) $\sum_{i=1}^N [\mathbf{f}_i - m_i \dot{\mathbf{V}}_i] \cdot \mathbf{v}_{ij} = 0$

where $Q_j = \sum_{i=1}^N \mathbf{f}_i \cdot \frac{\partial \mathbf{R}_i}{\partial \mathbf{q}_j} = \sum_{i=1}^N \mathbf{f}_i \cdot \mathbf{v}_{ij}$ and $\mathbf{v}_{ij} = \frac{\partial \mathbf{R}_i}{\partial \dot{\mathbf{q}}_j} = \frac{\partial \mathbf{R}_i}{\partial \dot{\mathbf{q}}_k}$.

We are now prepared to develop the differential equations of motion using d'Alembert's principle (particularly, from-1):

$$\begin{aligned} m_1 \ddot{\mathbf{R}}_1 \cdot \frac{\partial \mathbf{R}_1}{\partial x} + m_2 \ddot{\mathbf{R}}_2 \cdot \frac{\partial \mathbf{R}_2}{\partial x} &= Q_x \\ m_1 \ddot{\mathbf{R}}_1 \cdot \frac{\partial \mathbf{R}_1}{\partial \theta} + m_2 \ddot{\mathbf{R}}_2 \cdot \frac{\partial \mathbf{R}_2}{\partial \theta} &= Q_\theta \end{aligned}$$

After substituting everything we prepared, the first equation is

$$\begin{aligned} m_1 \ddot{x} \hat{\mathbf{n}}_1 \cdot \hat{\mathbf{n}}_1 + m_2 ((\ddot{x} - r\dot{\theta}^2 \sin \theta + r\ddot{\theta} \cos \theta) \hat{\mathbf{n}}_1 + (r\dot{\theta}^2 \cos \theta + r\ddot{\theta} \sin \theta) \hat{\mathbf{n}}_2) \cdot \hat{\mathbf{n}}_1 &= -kx \\ m_1 \ddot{x} + m_2 (\ddot{x} - r\dot{\theta}^2 \sin \theta + r\ddot{\theta} \cos \theta) &= -kx \\ (m_1 + m_2) \ddot{x} + m_2 r \ddot{\theta} \cos \theta - m_2 r \dot{\theta}^2 \sin \theta &= -kx \end{aligned}$$

and the second equation is

$$\begin{aligned} m_1 \ddot{x} \hat{\mathbf{n}}_1 \cdot \mathbf{0} + m_2 ((\ddot{x} - r\dot{\theta}^2 \sin \theta + r\ddot{\theta} \cos \theta) \hat{\mathbf{n}}_1 + (r\dot{\theta}^2 \cos \theta + r\ddot{\theta} \sin \theta) \hat{\mathbf{n}}_2) \cdot (r \cos \theta \hat{\mathbf{n}}_1 + r \sin \theta \hat{\mathbf{n}}_2) &= -m_2 g r \sin \theta \\ m_2 (\ddot{x} - r\dot{\theta}^2 \sin \theta + r\ddot{\theta} \cos \theta) r \cos \theta + m_2 (r\dot{\theta}^2 \cos \theta + r\ddot{\theta} \sin \theta) r \sin \theta &= -m_2 g r \sin \theta \\ m_2 r \cos \theta \ddot{x} + m_2 r^2 \ddot{\theta} &= -m_2 g r \sin \theta \\ \cos \theta \ddot{x} + r \ddot{\theta} &= -g \sin \theta \end{aligned}$$

Newtonian Dynamics with Constraints

The preceding developments implicitly assume that the generalized coordinates are independent. It often occurs that the coordinates are not independent. In the simplest case, the redundancy of the coordinates arises because of constraining algebraic relationships of the form

$$\psi_k(t, q_1, q_2, \dots, q_n) = 0 \quad k = 1, 2, \dots, m \quad (5.54)$$

which is referred to as **holonomic constraints**. If time does not explicitly appear in the constraint in Eq. (5.54), then this special case of holonomic constraints are said to be scleronomic.

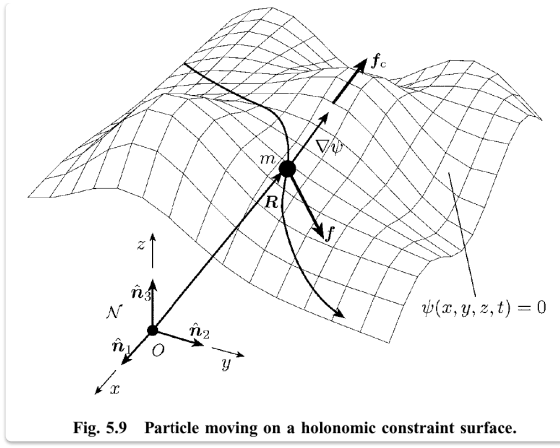
Some examples of **nonholonomic constraints**:

- Inequality constraints
- Velocity-dependent constraint that cannot be integrated to obtain this form

There are usually two obvious approaches to **dealing with the holonomic constraint**:

1. Solve the constraint equation for any one of the coordinates as a function of the other coordinates q_i that may be considered independent. Then, the d'Alembert's principle can be directly applied.
2. Replace the constraint surface by an equivalent constraint force that effectively causes the motion to remain on the constraint surface. Then, the constraint force has to be taken into consideration when applying the d'Alembert's principle, because the admissible virtual work is not confined on the constraint surface anymore.

✍ Under constraints, the d'Alembert's principle requires admissible/consistent virtual displacement to get rid of constraint forces explicitly. When the constraint forces need to be handled explicitly, Newtonian mechanics or Lagrange multipliers are the fallbacks.



For the case of introducing the constraint force $\mathbf{f}_c$ that is normal to the smooth holonomic constraint surface $\psi = 0$, as shown in Fig. 5.9,

$$\mathbf{f}_c = \lambda \frac{\partial \psi}{\partial x} \hat{\mathbf{n}}_1 + \lambda \frac{\partial \psi}{\partial y} \hat{\mathbf{n}}_2 + \lambda \frac{\partial \psi}{\partial z} \hat{\mathbf{n}}_3 \quad (5.68)$$

Adding this constraint force into Newton's second law, we are able to get differential equations with additionally constraints.

Newton's second law provides the equation of motion

$$\mathbf{f} + \mathbf{f}_c = m\ddot{\mathbf{R}} \quad (5.63)$$

which can be explicitly expressed as

$$\begin{aligned} m\ddot{x} &= f_x + \lambda \frac{\partial \psi}{\partial x} \\ m\ddot{y} &= f_y + \lambda \frac{\partial \psi}{\partial y} \\ m\ddot{z} &= f_z + \lambda \frac{\partial \psi}{\partial z} \end{aligned} \quad (5.69)$$

Together with a algebraic equation (the constraint function)

$$\psi(x, y, z, t) = 0 \quad (5.70)$$

From Eqs. (5.69) and (5.70), we observe that the constraint force $\mathbf{f}_c$ is not necessarily to be written out explicitly but what we need is just the function ψ and its partial derivatives.

Next, we observe another feature that enables an extension from holonomic constraints to a more general type of constraints. Differential change dx, dy, dz along the trajectory satisfies the following constraint:

$$\frac{d\psi}{dt} = \frac{\partial \psi}{\partial x} \dot{x} + \frac{\partial \psi}{\partial y} \dot{y} + \frac{\partial \psi}{\partial z} \dot{z} + \frac{\partial \psi}{\partial t} = 0$$

which is equivalent to

$$d\psi = \frac{\partial \psi}{\partial x} dx + \frac{\partial \psi}{\partial y} dy + \frac{\partial \psi}{\partial z} dz + \frac{\partial \psi}{\partial t} dt = 0. \quad (5.66)$$

Admissible virtual change $\delta x, \delta y, \delta z$ are along the constraint surface at a fixed moment without an increment in time t , which can be expressed as

$$\delta\psi = \frac{\partial \psi}{\partial x} \delta x + \frac{\partial \psi}{\partial y} \delta y + \frac{\partial \psi}{\partial z} \delta z = 0 \quad (5.67)$$

So, the essential of the constraint ψ is actually its differential form in Eq. (5.66).

Now, we introduce **Pfaffian Nonholonomic Constraints** which are a special generalization of holonomic constraints and can be expressed commonly as

$$B(x, y, z, t) + A_1(x, y, z, t)\dot{x} + A_2(x, y, z, t)\dot{y} + A_3(x, y, z, t)\dot{z} = 0 \quad (5.84)$$

or the differential format

$$Bdt + A_1dx + A_2dy + A_3dz = 0$$

Notice that it will degenerate to holonomic constraint if there is a function ψ whose partial derivatives are the coefficients B, A_1, A_2, A_3 above.

For the case of m constraints and n generalized coordinates, consider N particles with m Pfaffian nonholonomic constraints:

$$\sum_{j=1}^n A_{kj} \dot{q}_j + B_k = 0 \quad k = 1, 2, \dots, m \quad (5.99)$$

where $A_{kj} = A_{kj}(q_1, q_2, \dots, q_n, t)$ and $B_k = B_k(q_1, q_2, \dots, q_n, t)$ are functions of generalized coordinates q_j and time. This can also be expressed as

$$\sum_{j=1}^n A_{kj} dq_j + B_k dt = 0 \quad k = 1, 2, \dots, m \quad (5.100)$$

For the case that $\mathbf{q}$ is the set of $n = 3N$ Cartesian inertial coordinates, the equations of motion for the N particles are n second-order ODEs

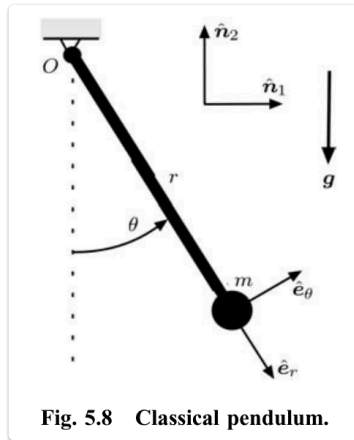
$$M_j \ddot{q}_j = f_j + f_{c_j} = f_j + \sum_{k=1}^m A_{kj} \lambda_k \quad j = 1, 2, \dots, n \quad (5.102)$$

and m first-order ODEs

$$\sum_{j=1}^n A_{kj} \dot{q}_j + B_k = 0 \quad k = 1, 2, \dots, m \quad (5.103)$$

The Eqs. (5.102) and (5.103) constitute a set of **differential-algebraic equations (DAEs)**.

Example 5.6: classical pendulum (two approaches to embed constraints in Derivations)



Algebraic constraint elimination:

$$\mathbf{R} = r \hat{e}_r, \quad \dot{\mathbf{R}} = \dot{r} \hat{e}_r + r \dot{\theta} \hat{e}_\theta, \quad \ddot{\mathbf{R}} = (\ddot{r} - r \dot{\theta}^2) \hat{e}_r + (r \ddot{\theta} + 2 \dot{r} \dot{\theta}) \hat{e}_\theta$$

and there is a holonomic constraint

$$\psi(r, \theta) = r - R = 0$$

so that we are only left with a single independent coordinate θ :

$$\mathbf{R} = R \hat{e}_r, \quad \dot{\mathbf{R}} = R \dot{\theta} \hat{e}_\theta, \quad \ddot{\mathbf{R}} = -(R \dot{\theta}^2) \hat{e}_r + (R \ddot{\theta}) \hat{e}_\theta$$

✓ Different forms of d'Alembert's principle:

1. (classical form, Eq. 5.24) $\sum_{i=1}^N m_i \ddot{\mathbf{R}}_i \cdot \frac{\partial \mathbf{R}_i}{\partial \mathbf{q}_j} = Q_j$
2. $\sum_{i=1}^N m_i \ddot{\mathbf{R}}_i \cdot \mathbf{v}_{ij} = Q_j$
3. (virtual power form, Eq. 5.30) $\sum_{i=1}^N [\mathbf{f}_i - m_i \dot{\mathbf{V}}_i] \cdot \mathbf{v}_{ij} = 0$

where $Q_j = \sum_{i=1}^N \mathbf{f}_i \cdot \frac{\partial \mathbf{R}_i}{\partial \mathbf{q}_j} = \sum_{i=1}^N \mathbf{f}_i \cdot \mathbf{v}_{ij}$ and $\mathbf{v}_{ij} = \frac{\partial \mathbf{R}_i}{\partial \dot{\mathbf{q}}_j} = \frac{\partial \mathbf{R}_i}{\partial \dot{\mathbf{q}}_k}$.

Recall the d'Alembert's principle above and apply the classical form here:

$$\begin{aligned}
 m\ddot{\mathbf{R}} \cdot \frac{\partial \mathbf{R}}{\partial \theta} &= Q_\theta \\
 m \left(-(R\dot{\theta}^2)\hat{\mathbf{e}}_r + (R\ddot{\theta})\hat{\mathbf{e}}_\theta \right) \cdot \frac{\partial}{\partial \theta} (R\hat{\mathbf{e}}_r) &= \mathbf{f} \cdot \frac{\partial \mathbf{R}}{\partial \theta} \\
 m \left(-(R\dot{\theta}^2)\hat{\mathbf{e}}_r + (R\ddot{\theta})\hat{\mathbf{e}}_\theta \right) \cdot (R\hat{\mathbf{e}}_\theta) &= mg(\cos \theta \hat{\mathbf{e}}_r - \sin \theta \hat{\mathbf{e}}_\theta) \cdot (R\hat{\mathbf{e}}_\theta) \\
 mR^2\ddot{\theta} &= -mgR \sin \theta \\
 \ddot{\theta} &= -\frac{g}{R} \sin \theta
 \end{aligned}$$

Constraint force via Lagrange multipliers: Thus the constraint force associated with $\psi(r, \theta) = r - R = 0$ is written as a centripetal force $\mathbf{F}_c$

$$\mathbf{F}_c = \lambda \nabla \psi(r, \theta) = \lambda \hat{\mathbf{e}}_r$$

The total force acting on mass m is

$$\mathbf{F} = -mg\hat{\mathbf{n}}_2 + \lambda \hat{\mathbf{e}}_r$$

and the d'Alembert's principle (the classical form) gives two equations this time

✓ **Different forms of d'Alembert's principle:**

1. (classical form, Eq. 5.24) $\sum_{i=1}^N m_i \ddot{\mathbf{R}}_i \cdot \frac{\partial \mathbf{R}_i}{\partial q_j} = Q_j$
2. $\sum_{i=1}^N m_i \ddot{\mathbf{R}}_i \cdot \mathbf{v}_{ij} = Q_j$
3. (virtual power form, Eq. 5.30) $\sum_{i=1}^N [\mathbf{f}_i - m_i \dot{\mathbf{V}}_i] \cdot \mathbf{v}_{ij} = 0$

where $Q_j = \sum_{i=1}^N \mathbf{f}_i \cdot \frac{\partial \mathbf{R}_i}{\partial q_j} = \sum_{i=1}^N \mathbf{f}_i \cdot \mathbf{v}_{ij}$ and $\mathbf{v}_{ij} = \frac{\partial \mathbf{R}_i}{\partial \dot{q}_j} = \frac{\partial \mathbf{R}_i}{\partial \dot{q}_k}$.

$$\begin{aligned}
 m\ddot{\mathbf{R}} \cdot \frac{\partial \mathbf{R}}{\partial r} &= Q_r \\
 m\ddot{\mathbf{R}} \cdot \frac{\partial \mathbf{R}}{\partial \theta} &= Q_\theta
 \end{aligned}$$

which leads to

$$\begin{aligned}
 m \left[(\ddot{r} - r\dot{\theta}^2)\hat{\mathbf{e}}_r + (r\ddot{\theta} + 2r\dot{\theta})\hat{\mathbf{e}}_\theta \right] \cdot \hat{\mathbf{e}}_r &= [-mg\hat{\mathbf{n}}_2 + \lambda \hat{\mathbf{e}}_r] \cdot \hat{\mathbf{e}}_r = mg \cos \theta + \lambda \\
 m \left[(\ddot{r} - r\dot{\theta}^2)\hat{\mathbf{e}}_r + (r\ddot{\theta} + 2r\dot{\theta})\hat{\mathbf{e}}_\theta \right] \cdot r\hat{\mathbf{e}}_\theta &= [-mg\hat{\mathbf{n}}_2 + \lambda \hat{\mathbf{e}}_r] \cdot (r\hat{\mathbf{e}}_\theta) = -mgr \sin \theta
 \end{aligned}$$

Because $r = R$ is a constant, $\dot{r} = \ddot{r} = 0$ and we have

$$\begin{aligned}
 \lambda &= -mg \cos \theta - mR\dot{\theta} \\
 \ddot{\theta} &= -\frac{g}{R} \sin \theta
 \end{aligned}$$

which is the same to the previous approach.

① **Partial derivatives in polar coordinates** $\frac{\partial \hat{\mathbf{e}}_r}{\partial \theta} = \hat{\mathbf{e}}_\theta$, $\frac{\partial \hat{\mathbf{e}}_\theta}{\partial \theta} = -\hat{\mathbf{e}}_r$. >

Proof #1: resolving to $\hat{\mathbf{n}}_1, \hat{\mathbf{n}}_2$ to show θ explicitly, then taking partial derivatives directly.

Proof #2:

$$d\hat{\mathbf{e}}_r = \frac{\partial \hat{\mathbf{e}}_r}{\partial t} dt + \frac{\partial \hat{\mathbf{e}}_r}{\partial r} dr + \frac{\partial \hat{\mathbf{e}}_r}{\partial \theta} d\theta \quad \text{(total derivative)}$$

$$\frac{\mathcal{N}d}{dt} \hat{\mathbf{e}}_r = \frac{\varepsilon d}{dt} \hat{\mathbf{e}}_r + \dot{\theta} \hat{\mathbf{e}}_3 \times \hat{\mathbf{e}}_r = \dot{\theta} \hat{\mathbf{e}}_\theta \quad \text{(transport theorem)}$$

and

$$d\hat{\mathbf{e}}_\theta = \frac{\partial \hat{\mathbf{e}}_\theta}{\partial t} dt + \frac{\partial \hat{\mathbf{e}}_\theta}{\partial r} dr + \frac{\partial \hat{\mathbf{e}}_\theta}{\partial \theta} d\theta \quad \text{(total derivative)}$$

$$\frac{\mathcal{N}d}{dt} \hat{\mathbf{e}}_\theta = \frac{\varepsilon d}{dt} \hat{\mathbf{e}}_\theta + \dot{\theta} \hat{\mathbf{e}}_3 \times \hat{\mathbf{e}}_\theta = -\dot{\theta} \hat{\mathbf{e}}_r \quad \text{(transport theorem)}$$

Compare the coefficients and we get the results.

D'Alembert's principle vs. Newton's second law

- A fundamental advantage over Newton's second law in that the internal forces and all other virtually nonworking constraint forces can be simply ignored in developing the equations of motion.
- The vector kinematic algebraic overhead associated with Newton's second law and d'Alembert's principle is essentially identical, because both require vector kinematics to be taken through the acceleration level.

Lagrangian Dynamics

- Lagrange's equations require only velocity-level vector kinematics.

Minimal Coordinate Systems and Unconstrained Motion

An assumption for the following derivation is that, if constraints are present, they are simple algebraic holonomic constraints that have been kinematically eliminated to establish a minimal coordinate description of the system, so the generalized coordinates must be independent.

✓ Different forms of d'Alembert's principle:

1. (classical form, Eq. 5.24) $\sum_{i=1}^N m_i \ddot{\mathbf{R}}_i \cdot \frac{\partial \mathbf{R}_i}{\partial \mathbf{q}_j} = Q_j$
2. $\sum_{i=1}^N m_i \ddot{\mathbf{R}}_i \cdot \mathbf{v}_{ij} = Q_j$
3. (virtual power form, Eq. 5.30) $\sum_{i=1}^N [\mathbf{f}_i - m_i \dot{\mathbf{V}}_i] \cdot \mathbf{v}_{ij} = 0$

where $Q_j = \sum_{i=1}^N \mathbf{f}_i \cdot \frac{\partial \mathbf{R}_i}{\partial \mathbf{q}_j} = \sum_{i=1}^N \mathbf{f}_i \cdot \mathbf{v}_{ij}$ and $\mathbf{v}_{ij} = \frac{\partial \dot{\mathbf{R}}_i}{\partial \dot{\mathbf{q}}_j} = \frac{\partial \mathbf{R}_i}{\partial \mathbf{q}_j}$.

Rewrite Eq. (5.24) explicitly as

$$\sum_{i=1}^N m_i \ddot{\mathbf{R}}_i \cdot \frac{\partial \mathbf{R}_i}{\partial \mathbf{q}_j} = \sum_{i=1}^N \mathbf{f}_i \cdot \frac{\partial \mathbf{R}_i}{\partial \mathbf{q}_j} \quad \text{for } j = 1, 2, \dots, n$$

and by plugging in the cancellation of dots identity, we have

$$\sum_{i=1}^N m_i \ddot{\mathbf{R}}_i \cdot \frac{\partial \mathbf{R}_i}{\partial \dot{\mathbf{q}}_j} = \sum_{i=1}^N \mathbf{f}_i \cdot \frac{\partial \dot{\mathbf{R}}_i}{\partial \dot{\mathbf{q}}_j} \quad \text{for } j = 1, 2, \dots, n \quad (5.135)$$

The kinetic energy $T(t, q_1, q_2, \dots, q_n; \dot{q}_1, \dot{q}_2, \dots, \dot{q}_n)$ is generally a function of both q_i and $\dot{q}_i$,

$$T = \frac{1}{2} \sum_{i=1}^N m_i \dot{\mathbf{R}}_i \cdot \dot{\mathbf{R}}_i \quad (5.136)$$

so the partial derivatives with respect to $(q_j, \dot{q}_j)$ are

$$\begin{aligned} \frac{\partial T}{\partial q_j} &= \sum_{i=1}^N m_i \dot{\mathbf{R}}_i \cdot \frac{\partial \dot{\mathbf{R}}_i}{\partial q_j} \\ \frac{\partial T}{\partial \dot{q}_j} &= \sum_{i=1}^N m_i \dot{\mathbf{R}}_i \cdot \frac{\partial \dot{\mathbf{R}}_i}{\partial \dot{q}_j} \end{aligned} \quad (5.137)$$

Now we can do

$$\begin{aligned}
\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) &= \frac{d}{dt} \left(\sum_{i=1}^N m_i \dot{\mathbf{R}}_i \cdot \frac{\partial \dot{\mathbf{R}}_i}{\partial \dot{q}_j} \right) && \text{(used 5.137)} \\
&= \sum_{i=1}^N m_i \ddot{\mathbf{R}}_i \cdot \frac{\partial \dot{\mathbf{R}}_i}{\partial \dot{q}_j} + \sum_{i=1}^N m_i \dot{\mathbf{R}}_i \cdot \frac{d}{dt} \left(\frac{\partial \dot{\mathbf{R}}_i}{\partial \dot{q}_j} \right) \\
&= \sum_{i=1}^N (\mathbf{f}_i + \mathbf{f}_{ci}) \cdot \frac{\partial \dot{\mathbf{R}}_i}{\partial \dot{q}_j} + \sum_{i=1}^N m_i \dot{\mathbf{R}}_i \cdot \frac{d}{dt} \left(\frac{\partial \dot{\mathbf{R}}_i}{\partial \dot{q}_j} \right) \\
&= \sum_{i=1}^N \mathbf{f}_i \cdot \frac{\partial \dot{\mathbf{R}}_i}{\partial \dot{q}_j} + \sum_{i=1}^N \mathbf{f}_{ci} \cdot \frac{\partial \dot{\mathbf{R}}_i}{\partial \dot{q}_j} + \sum_{i=1}^N m_i \dot{\mathbf{R}}_i \cdot \frac{d}{dt} \left(\frac{\partial \dot{\mathbf{R}}_i}{\partial \dot{q}_j} \right) && \text{(cancel dots for last t)} \\
&= \sum_{i=1}^N \mathbf{f}_i \cdot \frac{\partial \dot{\mathbf{R}}_i}{\partial \dot{q}_j} + \sum_{i=1}^N \mathbf{f}_{ci} \cdot \sum_{j=1}^n \frac{\partial \dot{\mathbf{R}}_i}{\partial q_j} \delta q_j + \sum_{i=1}^N m_i \dot{\mathbf{R}}_i \cdot \frac{d}{dt} \left(\frac{\partial \dot{\mathbf{R}}_i}{\partial \dot{q}_j} \right) && \text{(special admissible } \delta \mathbf{R}_i \text{ with } \delta \dot{\mathbf{R}}_i = \sum_{j=1}^n \frac{\partial \dot{\mathbf{R}}_i}{\partial q_j} \delta q_j) \\
&= \sum_{i=1}^N \mathbf{f}_i \cdot \frac{\partial \dot{\mathbf{R}}_i}{\partial \dot{q}_j} + \sum_{i=1}^N m_i \dot{\mathbf{R}}_i \cdot \frac{\partial}{\partial q_j} \left(\frac{d}{dt} \dot{\mathbf{R}}_i \right) && \text{(smooth } \mathbf{R}_i, \text{ switch } \frac{d}{dt} \text{ to } \frac{d}{dt} \text{)} \\
&= \sum_{i=1}^N \mathbf{f}_i \cdot \frac{\partial \dot{\mathbf{R}}_i}{\partial \dot{q}_j} + \frac{\partial T}{\partial q_j} && \text{(used 5.137)} \\
&= Q_j + \frac{\partial T}{\partial q_j}
\end{aligned}$$

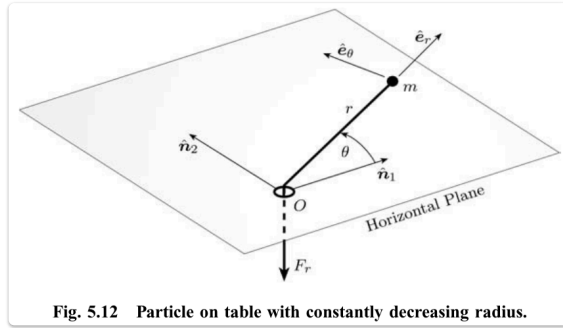
Therefore, we have the **most fundamental version of Lagrange's equations**,

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} = \sum_{i=1}^N \mathbf{f}_i \cdot \frac{\partial \dot{\mathbf{R}}_i}{\partial \dot{q}_j} \equiv Q_j \quad \text{for } j = 1, 2, \dots, n \quad (5.139)$$

An appropriate definition of kinetic energy results in these equations applying to systems of rigid bodies and particles.

Again, the assumption is that the the generalized coordinates $\{q_1, q_2, \dots, q_n\}$ must be independent in Eq. (5.139).

Example 5.10: particles with decreasing radius (Applying Lagrange's equations)



A particle moves on a table. A string attached to the particle is being drawn through a hole such that the radius is decreasing at a constant rate; determine angular velocity as a function of time.

A reference frame $\{\hat{n}\}$ and a polar frame $\{\hat{e}_r, \hat{e}_\theta\}$.

The radial speed is a constant $v_r = \dot{r} = -c$, where $c > 0$ is a constant.

The only generalized coordinate is θ .

The position vector is $\mathbf{R} = r\hat{e}_r$.

The velocity vector is $\dot{\mathbf{R}} = \dot{r}\hat{e}_r + r\dot{\theta}\hat{e}_\theta = -c\hat{e}_r + r\dot{\theta}\hat{e}_\theta$

The kinetic energy is $T = \frac{1}{2}m(c^2 + r^2\dot{\theta}^2)$

Apply the Lagrange's equation:

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{\theta}} \right) - \frac{\partial T}{\partial \theta} = \sum_{i=1}^N \mathbf{f}_i \cdot \frac{\partial \dot{\mathbf{R}}_i}{\partial \dot{\theta}} \equiv Q_j \quad \text{for } j = 1, 2, \dots, n \quad (5.139)$$

and we have

$$\begin{aligned}
\frac{d}{dt} (mr^2\dot{\theta}) - 0 &= -F_r \hat{e}_r \cdot (r\hat{e}_\theta) \\
\frac{d}{dt} (mr^2\dot{\theta}) &= 0
\end{aligned}$$

Since $r = r_0 - ct$ is time-varying, we have

$$(r_0 - ct)^2 \dot{\theta} = r_0^2 \dot{\theta}_0 \implies \dot{\theta} = \frac{r_0^2}{(r_0 - ct)^2} \dot{\theta}_0$$

Lagrange's Equations for Conservative Forces

For the conservative force $\mathbf{f}_i$ determined by the potential function $V(t, q_1, q_2, \dots, q_n)$, we have

$$\mathbf{f}_i = -\frac{\partial V}{\partial \mathbf{R}_i}$$

Then the generalized force is given by

$$Q_j = \sum_{i=1}^N \mathbf{f}_i \cdot \frac{\partial \mathbf{R}_i}{\partial q_j} = - \sum_{i=1}^N \frac{\partial V}{\partial \mathbf{R}_i} \cdot \frac{\partial \mathbf{R}_i}{\partial q_j} = - \frac{\partial V}{\partial q_j} \quad (5.151)$$

Define the **Lagrangian function** $\mathcal{L}$ as

$$\mathcal{L} = \mathcal{L}(t, q_1, q_2, \dots, q_n; \dot{q}_1, \dot{q}_2, \dots, \dot{q}_n) \equiv T - V \quad (5.152)$$

so that we have

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \dot{q}_j} &= \frac{\partial T}{\partial \dot{q}_j} \\ \frac{\partial \mathcal{L}}{\partial q_j} &= \frac{\partial T}{\partial q_j} - \frac{\partial V}{\partial q_j} \end{aligned}$$

If all forces are conservative, for the fundamental format of Lagrange's equations

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} = \sum_{i=1}^N \mathbf{f}_i \cdot \frac{\partial \dot{\mathbf{R}}_i}{\partial \dot{q}_j} \equiv Q_j \quad \text{for } j = 1, 2, \dots, n \quad (5.139)$$

we can move all Q_j to LHS and plugin Eq. (5.151) to get

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} + \frac{\partial V}{\partial q_j} = 0$$

Then, plug in the definition of $\mathcal{L}$ and its partial derivatives, we get the most famous form of Lagrange's equations, in the case when all forces are conservative,

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right) - \frac{\partial \mathcal{L}}{\partial q_j} = 0 \quad (5.153)$$

① For many elementary conservative systems, the potential and kinetic energy can be simply written with a minimum of derivations; for these cases, Eq. (5.153) do not require derivation of any generalized forces and are therefore especially attractive.

For more general cases with both conservative and nonconservative forces, we just need to retain the generalized forces at the RHS, resulting,

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right) - \frac{\partial \mathcal{L}}{\partial q_j} = Q_{nc_j} \quad (5.154)$$

where the nonconservative generalized force $Q_{nc_j} = \sum_{i=1}^N \mathbf{f}_{nc_i} \cdot \frac{\partial \dot{\mathbf{R}}_i}{\partial \dot{q}_j}$.

Example 5.12: linear spring pendulum (Applying Lagrange's equations)

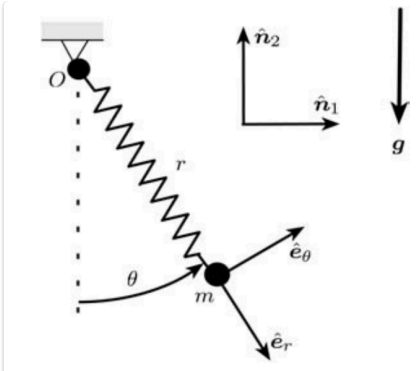


Fig. 5.14 Classical spring pendulum.

the linear spring pendulum is considered: nominal unstretched spring length r_0 , linear spring constant k . Develop the equations of motion.

The only virtually working forces are the spring force and gravity, and that there are two generalized coordinates $\{r, \theta\}$.

The position vector is $\mathbf{R} = r\hat{e}_r$,

The velocity vector is $\mathbf{V} = \dot{\mathbf{R}} = \dot{r}\hat{e}_r + r\dot{\theta}\hat{e}_\theta$

The kinetic energy is $T = \frac{1}{2}m\mathbf{V} \cdot \mathbf{V} = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2)$

Both the gravity and the spring force are conservative, so the total potential function is $V = -mgr \cos \theta + \frac{1}{2}k(r - r_0)^2$

So the Lagrangian function is $\mathcal{L} = T - V = \frac{1}{2}m\mathbf{V} \cdot \mathbf{V} = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) + mgr \cos \theta - \frac{1}{2}k(r - r_0)^2$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right) - \frac{\partial \mathcal{L}}{\partial q_j} = 0 \quad (5.153)$$

Applying the Lagrange's equations for conservative forces, we have

$$\begin{aligned} \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{r}} \right) - \frac{\partial \mathcal{L}}{\partial r} &= 0 \\ \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}} \right) - \frac{\partial \mathcal{L}}{\partial \theta} &= 0 \end{aligned}$$

which gives

$$\begin{aligned} \frac{d}{dt} (m\dot{r}) - m\dot{\theta}^2 - mg \cos \theta + k(r - r_0) &= 0 \\ \frac{d}{dt} (mr^2\dot{\theta}) - mgr \sin \theta &= 0 \end{aligned}$$

then

$$\begin{aligned} m\ddot{r} - m\dot{\theta}^2 - mg \cos \theta + k(r - r_0) &= 0 \\ 2mr\dot{r}\dot{\theta} + mr^2\ddot{\theta} + mgr \sin \theta &= 0 \end{aligned}$$

Alternatively, if we want to apply the other format of Lagrange's equations

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} = \sum_{i=1}^N \mathbf{f}_i \cdot \frac{\partial \dot{\mathbf{R}}_i}{\partial \dot{q}_j} \equiv Q_j \quad \text{for } j = 1, 2, \dots, n \quad (5.139)$$

we need to first calculate the two generalized forces as

$$\begin{aligned} Q_r &= \left(-k(r - r_0)\hat{e}_r + mg \cos \theta \hat{e}_r - mg \sin \theta \hat{e}_\theta \right) \cdot \frac{\partial (\dot{r}\hat{e}_r + r\dot{\theta}\hat{e}_\theta)}{\partial \dot{r}} \\ Q_\theta &= \left(-k(r - r_0)\hat{e}_r + mg \cos \theta \hat{e}_r - mg \sin \theta \hat{e}_\theta \right) \cdot \frac{\partial (\dot{r}\hat{e}_r + r\dot{\theta}\hat{e}_\theta)}{\partial \dot{\theta}} \end{aligned}$$

Obviously we will get the same results but with a bit more steps and algebras. These force has been implicitly accounted for by being included in the potential energy function.

Problem 5.1: linear torsional spring (comparing different methods to obtain EOMs)

- 5.1 Consider the system shown in Fig. P5.1. The pendulum has a weightless rod of length R . In addition to gravity, consider the torsional linear spring (moment $= -k\theta$, spring potential energy $= \frac{1}{2}k\theta^2$). Your tasks are to derive the equations of motion using:
- Newton's laws.
 - Euler's equations of motion ($\mathbf{L} = \dot{\mathbf{H}}$), taking moment about pivot point O .
 - A version of Lagrange's equations.

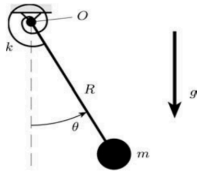
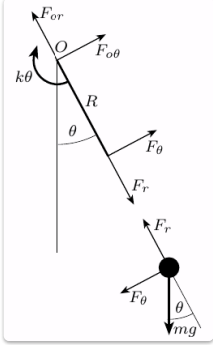


Fig. P5.1 Illustration of a pendulum with a linear torsional spring.

Use Newton's law and free-body diagrams for the rod and mass.



Summing the forces and torques on the massless rod: $0 \cdot \frac{d^2}{dt^2} \left(\frac{R}{2} \hat{e}_r \right) = (F_r - F_{or}) \hat{e}_r + (F_\theta + F_{o\theta}) \hat{e}_\theta$

$$-k\theta + R F_\theta = 0$$

from which we can resolve: $F_{or} = F_r$ $F_{o\theta} = -F_\theta$ $F_\theta = \frac{k\theta}{R}$

Summing the forces on the mass (no torques because it's treated as a mass point now):

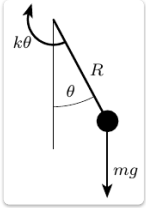
$$m(-R\ddot{\theta}^2 \hat{e}_r + R\ddot{\theta} \hat{e}_\theta) = (-F_r + mg \cos \theta) \hat{e}_r + (-F_\theta - mg \sin \theta) \hat{e}_\theta$$

$$F_r = mg \cos \theta + mR\ddot{\theta}^2$$

$$mR\ddot{\theta} = -\frac{k}{R}\theta - mg \sin \theta$$

And finally we have the equation of motion as a function of just the angle θ : $\ddot{\theta} = -\frac{k}{mR^2}\theta - \frac{g}{R}\sin \theta$

Using Euler's equations of rotational motion, which is essentially still Newton's law for special cases of rotation.



Apply the Euler's equation $\dot{\mathbf{H}} = \mathbf{L}$ directly to the total angular momentum (respect to O) $H = mR^2\dot{\theta}$. Notice it is a planar case so only the magnitude will be considered.

The torque relative to the pivot O can be easily formulated as $L = -mgR \sin \theta - k\theta$, where we assuming the positive direction for $\mathbf{L}$ is to point out of the screen/paper.

Then, we have $mR^2\ddot{\theta} = -k\theta - mgR \sin \theta$ which is already the EOM.

Using Lagrange's equations for conservative forces.

The generalized coordinate is just θ as apparently from previous free-body diagrams. The kinematic and potential functions are given as

$$T = \frac{1}{2} m \dot{\mathbf{R}} \cdot \dot{\mathbf{R}} = \frac{1}{2} m r^2 \dot{\theta}^2$$

$$V = mgR(1 - \cos \theta) + \frac{1}{2} k \theta^2$$

So, the Lagrangian function is $\mathcal{L} = T - V = \frac{1}{2} m R^2 \dot{\theta}^2 + mgR \cos \theta - \frac{1}{2} k \theta^2 - mgR$

Applying Lagrange's equations $\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}} \right) - \frac{\partial \mathcal{L}}{\partial q} = 0$ requires the following derivatives:

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}} \right) = mR^2 \ddot{\theta}$$

$$\frac{\partial \mathcal{L}}{\partial \theta} = -mgR \sin \theta - k\theta$$

Therefore the equation of motion is obtained as: $mR^2\ddot{\theta} + mgR \sin \theta + k\theta = 0$

❓ After-class practice, can the spring force be considered directly in the Lagrange's equations?

[Hint] Recall how the potential function V is absorbed into the Lagrange function $\mathcal{L}$.

✓ In this case, Euler's equation of rotational motion turns out to be the easiest approach. But clearly Lagrange's equations have provided a more systematic and streamlined approach.

Example 5.13: Generalize Example 5.3 with a damper (Applying Lagrange's equations with non-conservative forces)

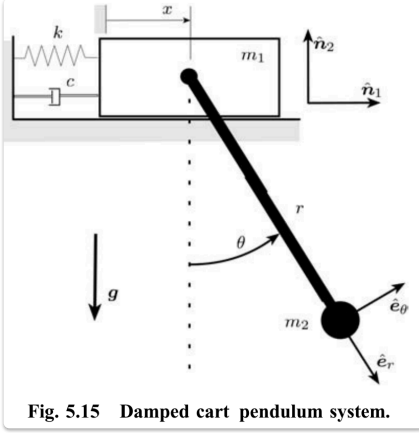


Fig. 5.15 Damped cart pendulum system.

The spring is accompanied with a dashpot linear damper with the damping force given as $-c\dot{x}\hat{n}_1$.

The generalized coordinates are still $\{x, \theta\}$.

The kinematics are

$$\dot{\mathbf{R}}_1 = \dot{x}\hat{n}_1 \quad \dot{\mathbf{R}}_2 = \dot{x}\hat{n}_1 + r\dot{\theta}\hat{e}_\theta$$

The virtually working force is $\mathbf{f}_d = -c\dot{x}\hat{n}_1$

The two generalized forces are (there are two generalized forces because of two generalized coordinates),

$$Q_x = \mathbf{f}_d \cdot \frac{\partial \dot{\mathbf{R}}_1}{\partial \dot{x}} + \cancel{\mathbf{f}_2 \cdot \frac{\partial \dot{\mathbf{R}}_2}{\partial \dot{x}}} = -c\dot{x}$$

$$Q_\theta = \mathbf{f}_d \cdot \frac{\partial \dot{\mathbf{R}}_1}{\partial \dot{\theta}} + \cancel{\mathbf{f}_2 \cdot \frac{\partial \dot{\mathbf{R}}_2}{\partial \dot{\theta}}} = 0$$

So the kinetic energy is $T = \frac{1}{2}(m_1\dot{\mathbf{R}}_1 \cdot \dot{\mathbf{R}}_1 + m_2\dot{\mathbf{R}}_2 \cdot \dot{\mathbf{R}}_2) = \frac{1}{2}(m_1\dot{x}^2 + m_2\dot{x}^2 + m_2r^2\dot{\theta}^2 + 2m_2\dot{x}r\dot{\theta}\cos\theta)$

and the potential energy function is $V = \frac{1}{2}kx^2 - m_2gr\cos\theta$

Thus, the Lagrangian function is $\mathcal{L} = T - V = \frac{1}{2}(m_1\dot{x}^2 + m_2\dot{x}^2 + m_2r^2\dot{\theta}^2 + 2m_2\dot{x}r\dot{\theta}\cos\theta) - \frac{1}{2}kx^2 + m_2gr\cos\theta$

Apply the Lagrange's equations with nonconservative forces below

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j}\right) - \frac{\partial \mathcal{L}}{\partial q_j} = Q_{nc_j} \quad (5.154)$$

we get

$$\frac{d}{dt}(m_1\dot{x} + m_2\dot{x} + m_2r\dot{\theta}\cos\theta) + kx = -c\dot{x}$$

$$\frac{d}{dt}(m_2r^2\dot{\theta} + m_2\dot{x}r\cos\theta) + m_2\dot{x}r\dot{\theta}\sin\theta + m_2gr\sin\theta = 0$$

After working out the derivatives, we get the EOMs as

$$m_1\ddot{x} + m_2\ddot{x} + m_2r\ddot{\theta}\cos\theta - m_2r\dot{\theta}^2\sin\theta + kx = -c\dot{x}$$

$$m_2r^2\ddot{\theta} + m_2\ddot{x}r\cos\theta - m_2\dot{x}r\dot{\theta}\sin\theta + m_2\dot{x}r\dot{\theta}\sin\theta + m_2gr\sin\theta = 0$$

Notice that we also don't need to calculate $\ddot{\mathbf{R}}_i$ in this approach, even though we need to calculate two "forces".

Redundant Coordinate Systems and Constrained Motion

Goal: Extend Lagrange's equations to consider redundant coordinates subject to Pfaffian nonholonomic constraints (which is more general and includes holonomic constraints).

Previously we assumed n was the number of degrees of freedom (which implicitly means all holonomic constraints have been eliminated, and that $\{q_1, q_2, \dots, q_n\}$ are a minimal set of independent coordinates. Under this case, the virtual displacement δq_j can be chosen arbitrarily. This can be expressed as the following principle that the virtual work is always zero for arbitrary δq_j

$$\delta W = \sum_{j=1}^n \left[Q_j + \frac{\partial T}{\partial q_j} - \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) \right] \delta q_j = 0 \quad (5.165)$$

Now, let's assume $\{q_1, q_2, \dots, q_n\}$ are not independent and several constraints are present. Furthermore, let's consider there are m Pfaffian form constraints

$$\sum_{j=1}^n A_{kj} \dot{q}_j + B_k = 0 \quad k = 1, 2, \dots, m \quad (5.166)$$

For instantaneous admissible virtual displacements under these constraints, we have

$$\sum_{j=1}^n A_{kj} \delta q_j = 0 \quad k = 1, 2, \dots, m \quad (5.168)$$

where $A_{kj} = A_{kj}(q_1, q_2, \dots, q_n, t)$ and $B_k = B_k(q_1, q_2, \dots, q_n, t)$ are functions of generalized coordinates q_j and time.

Analogous to the development of the constraint optimization Lagrange multiplier rule, we have

$$\delta W = \sum_{j=1}^n \left[Q_j + \sum_{k=1}^m \lambda_k A_{kj} + \frac{\partial T}{\partial q_j} - \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) \right] \delta q_j = 0 \quad (5.169)$$

Finally, we get the constrained version of Lagrange's equations of motion

$$\begin{aligned} \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} &= Q_j + \sum_{k=1}^m \lambda_k A_{kj} & j &= 1, 2, \dots, n \\ \sum_{j=1}^n A_{kj} \dot{q}_j + B_k &= 0 & k &= 1, 2, \dots, m \end{aligned} \quad (5.170 \text{ and } 171)$$

More conveniently, if there are some conservative forces, we can define the total potential function V and the Lagrangian function $\mathcal{L} = T - V$, then we get

$$\begin{aligned} \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right) - \frac{\partial \mathcal{L}}{\partial q_j} &= Q_{ncj} + \sum_{k=1}^m \lambda_k A_{kj} & j &= 1, 2, \dots, n \\ \sum_{j=1}^n A_{kj} \dot{q}_j + B_k &= 0 & k &= 1, 2, \dots, m \end{aligned} \quad (5.172)$$

Summarize of All Formats

✓ Summary of different forms of Lagrange's equation:

Fundamental with a minimum set of generalized coordinates:

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} = \sum_{i=1}^N \mathbf{f}_i \cdot \frac{\partial \dot{\mathbf{R}}_i}{\partial \dot{q}_j} \equiv Q_j \quad \text{for } j = 1, 2, \dots, n \quad (5.139)$$

Only conserved forces:

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right) - \frac{\partial \mathcal{L}}{\partial q_j} = 0 \quad (5.153)$$

Both conservative and nonconservative forces:

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right) - \frac{\partial \mathcal{L}}{\partial q_j} = Q_{ncj} \quad (5.154)$$

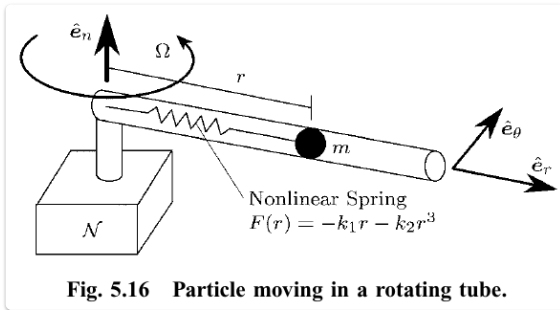
Redundant generalized coordinates with constraints:

$$\begin{aligned} \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} &= Q_j + \sum_{k=1}^m \lambda_k A_{kj} & j &= 1, 2, \dots, n \\ \sum_{j=1}^n A_{kj} \dot{q}_j + B_k &= 0 & k &= 1, 2, \dots, m \end{aligned} \quad (5.170 \text{ and } 171)$$

Ultimate version with everything:

$$\begin{aligned} \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right) - \frac{\partial \mathcal{L}}{\partial q_j} &= Q_{nc_j} + \sum_{k=1}^m \lambda_k A_{kj} & j &= 1, 2, \dots, n \\ \sum_{j=1}^n A_{kj} \dot{q}_j + B_k &= 0 & k &= 1, 2, \dots, m \end{aligned} \quad (5.172)$$

Example 5.14: Particle + Tube + Spring



The particle sliding in a rotating tube with a constant rotation speed of Ω . The spring force is given in the figure.

Generalized coordinate is just r .

Nonlinear spring force is a conserved force with $V = \int_0^r F(\rho) d\rho$

The kinematics are

$$\mathbf{R} = r\hat{\mathbf{e}}_r \quad \dot{\mathbf{R}} = \dot{r}\hat{\mathbf{e}}_r + r\dot{\theta}\hat{\mathbf{e}}_\theta = \dot{r}\hat{\mathbf{e}}_r + \dot{r}\Omega\hat{\mathbf{e}}_\theta$$

The Lagrangian function is obtained as

$$\begin{aligned} T &= \frac{1}{2} \dot{\mathbf{R}} \cdot \dot{\mathbf{R}} = \frac{1}{2} m (\dot{r}^2 + r^2 \Omega^2) \\ V &= \frac{1}{2} k_1 r^2 + \frac{1}{4} k_2 r^4 \\ \mathcal{L} &= T - V \end{aligned}$$

Applying the following format

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right) - \frac{\partial \mathcal{L}}{\partial q_j} = 0 \quad (5.153)$$

and we have

$$\frac{d}{dt} (m\dot{r}) - mr\Omega^2 + k_1 r + k_2 r^3 = 0$$

🔗 **After-class exercise:** Solve for the constraint force F_θ acting on the particle m . (Hint: use either Newton's second law or Lagrange's equation with a constraint $\dot{\theta} = \Omega$.)

Problem 5.4 (partial): Two particles connected by an ideal spring

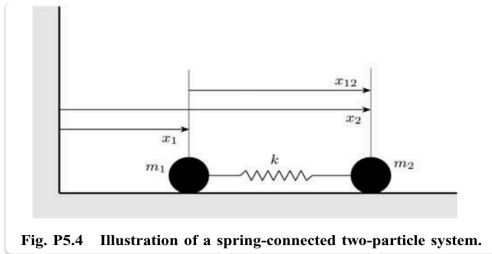


Fig. P5.4 Illustration of a spring-connected two-particle system.

Consider the two-particle system shown in Fig. P5.4. The particles move along a straight line on a frictionless plane. The unstretched spring length is d .

Consider the generalized coordinates $(q_1, q_2) \equiv (x_1, x_2)$, as shown in the figure, formulate the equations of motion using Lagrange's equations.

The force acting on m_1 is $k[(x_2 - x_1) - d] = k(x_{12} - d)$.

$$T = \frac{1}{2} (m_1 \dot{x}_1^2 + m_2 \dot{x}_2^2)$$

$$V = \frac{1}{2} k(x_{12} - d)^2 = \frac{1}{2} k(x_2 - x_1 - d)^2$$

$$\mathcal{L} = T - V = \frac{1}{2} (m_1 \dot{x}_1^2 + m_2 \dot{x}_2^2) - \frac{1}{2} k(x_2 - x_1 - d)^2$$

$$\frac{\partial \mathcal{L}}{\partial \dot{x}_1} = m_1 \dot{x}_1, \quad \frac{\partial \mathcal{L}}{\partial \dot{x}_2} = m_2 \dot{x}_2$$

$$\frac{\partial \mathcal{L}}{\partial x_1} = k(x_2 - x_1 - d), \quad \frac{\partial \mathcal{L}}{\partial x_2} = -k(x_2 - x_1 - d)$$

Apply Lagrange's equations for conservative forces:

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right) - \frac{\partial \mathcal{L}}{\partial q_j} = 0 \quad (5.153)$$

and we have

$$\begin{cases} m_1 \ddot{x}_1 - k(x_2 - x_1 - d) = 0 \\ m_2 \ddot{x}_2 + k(x_2 - x_1 - d) = 0 \end{cases}$$

🔗 After-class practice, obtain the equations of motion using a different set of generalized coordinate (x_c, x_{12}) , where x_c is the center of mass and x_{12} is the relative position of m_2 respect to m_1 .

[hint] $\ddot{x}_c = 0$, which can be directly verified using the results from (x_1, x_2) .

Example: Two particles with spring and dashpot damper (Problem 5.4 altered)

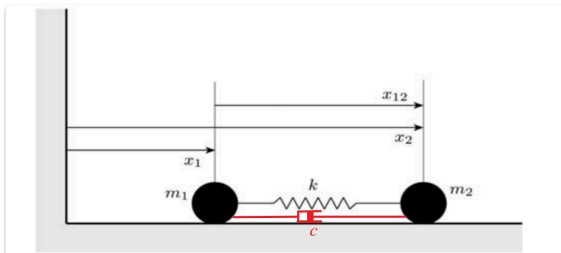


Fig. P5.4 Illustration of a spring-connected two-particle system.

Now, let's add a dashpot damper to the previous example.

$$T = \frac{1}{2} (m_1 \dot{x}_1^2 + m_2 \dot{x}_2^2)$$

$$V = \frac{1}{2} k(x_{12} - d)^2 = \frac{1}{2} k(x_2 - x_1 - d)^2$$

$$\mathcal{L} = T - V = \frac{1}{2} (m_1 \dot{x}_1^2 + m_2 \dot{x}_2^2) - \frac{1}{2} k(x_2 - x_1 - d)^2$$

$$\frac{\partial \mathcal{L}}{\partial \dot{x}_1} = m_1 \dot{x}_1, \quad \frac{\partial \mathcal{L}}{\partial \dot{x}_2} = m_2 \dot{x}_2$$

$$\frac{\partial \mathcal{L}}{\partial x_1} = k(x_2 - x_1 - d), \quad \frac{\partial \mathcal{L}}{\partial x_2} = -k(x_2 - x_1 - d)$$

Generalized forces from Non-conservative damping force $F = -c(\dot{x}_2 - \dot{x}_1)$ $Q_{x_1} = -[-c(\dot{x}_2 - \dot{x}_1)] \cdot \frac{\partial x_1}{\partial x_1} + [-c(\dot{x}_2 - \dot{x}_1)] \cdot \frac{\partial x_2}{\partial x_1} = c(\dot{x}_2 - \dot{x}_1)$

Similarly,

$$Q_{x_2} = -c(\dot{x}_2 - \dot{x}_1) \cdot \frac{\partial x_2}{\partial x_2} = -c(\dot{x}_2 - \dot{x}_1)$$

Apply Lagrange's equations for non-conservative forces:

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right) - \frac{\partial \mathcal{L}}{\partial q_j} = Q_{ncj} \quad (5.154)$$

and we have

$$\begin{cases} m_1 \ddot{x}_1 - k(x_2 - x_1 - d) = c(\dot{x}_2 - \dot{x}_1) \\ m_2 \ddot{x}_2 + k(x_2 - x_1 - d) = -c(\dot{x}_2 - \dot{x}_1) \end{cases}$$

① Extension to Lagrange's equations using "Rayleigh dissipation function".

Extensions to Torques and Rigid-body Dynamics

① This is covered in Example 5.5: Generalize force for a torque on rigid body

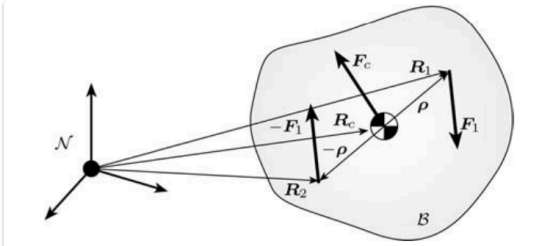


Fig. 5.7 Illustration of a torque being applied to a rigid body.

Both a force and a torque are being applied to a rigid body $\mathcal{B}$. The center of mass is $\mathbf{R}_c$ and is experiencing an external force $\mathbf{F}_c$. Simultaneously, the body is experiencing a torque $\mathbf{L}$.

A trivial trick we used here is to represent this torque as two forces centered at $\mathbf{R}_c$ and of equal magnitudes $|\mathbf{F}_1|$ that are acting in the opposite direction. Denote the relative vector of the exerting point as $\boldsymbol{\rho}$ so the total torque $\mathbf{L}$ is expressed as

$$\mathbf{L} = 2\boldsymbol{\rho} \times \mathbf{F}_1$$

According to the definition of the generalized force below

$$Q_j \equiv \sum_{i=1}^N \left(\mathbf{f}_i \cdot \frac{\partial \mathbf{R}_i}{\partial q_j} \right) \quad (5.22)$$

it is calculated as

$$\begin{aligned} Q_j &= \mathbf{F}_c \cdot \frac{\partial \mathbf{R}_c}{\partial q_j} + \mathbf{F}_1 \cdot \frac{\partial \mathbf{R}_1}{\partial q_j} + (-\mathbf{F}_1) \cdot \frac{\partial \mathbf{R}_2}{\partial q_j} \\ &= \mathbf{F}_c \cdot \frac{\partial \mathbf{R}_c}{\partial q_j} + \mathbf{F}_1 \cdot \frac{\partial (\mathbf{R}_c + \boldsymbol{\rho})}{\partial q_j} + (-\mathbf{F}_1) \cdot \frac{\partial (\mathbf{R}_c - \boldsymbol{\rho})}{\partial q_j} \\ &= \mathbf{F}_c \cdot \frac{\partial \mathbf{R}_c}{\partial q_j} + 2\mathbf{F}_1 \cdot \frac{\partial \boldsymbol{\rho}}{\partial q_j} \\ &= \mathbf{F}_c \cdot \frac{\partial \mathbf{R}_c}{\partial q_j} + 2\mathbf{F}_1 \cdot \frac{\partial \dot{\boldsymbol{\rho}}}{\partial \dot{q}_j} && \text{(identity)} \\ &= \mathbf{F}_c \cdot \frac{\partial \mathbf{R}_c}{\partial q_j} + 2\mathbf{F}_1 \cdot \frac{\partial (\boldsymbol{\omega} \times \boldsymbol{\rho})}{\partial \dot{q}_j} && \text{(rigid body)} \\ &= \mathbf{F}_c \cdot \frac{\partial \mathbf{R}_c}{\partial q_j} + 2\mathbf{F}_1 \cdot \left(\frac{\partial \boldsymbol{\omega}}{\partial \dot{q}_j} \times \boldsymbol{\rho} + \boldsymbol{\omega} \times \frac{\partial \boldsymbol{\rho}}{\partial \dot{q}_j} \right) && \text{(position vector)} \\ &= \mathbf{F}_c \cdot \frac{\partial \mathbf{R}_c}{\partial q_j} + \frac{\partial \boldsymbol{\omega}}{\partial \dot{q}_j} \cdot (\boldsymbol{\rho} \times 2\mathbf{F}_1) && \text{(triple product)} \\ &= \mathbf{F}_c \cdot \frac{\partial \mathbf{R}_c}{\partial q_j} + \mathbf{L} \cdot \frac{\partial \boldsymbol{\omega}}{\partial \dot{q}_j} \end{aligned}$$

For the generalized coordinate q_j , the corresponding generalized force Q_j is

$$Q_j = \mathbf{F}_c \cdot \frac{\partial \mathbf{R}_c}{\partial q_j} + \mathbf{L} \cdot \frac{\partial \boldsymbol{\omega}}{\partial \dot{q}_j} = \mathbf{F}_c \cdot \frac{\partial \mathbf{R}_c}{\partial q_j} + \mathbf{L} \cdot \frac{\partial \boldsymbol{\omega}}{\partial \dot{q}_j} \quad (5.53)$$

As a summary, we have dealt with a force as a pair of forces, then the assumed forces and displacements cancel out as the torque itself at the end. It is not surprising since Euler's equation is just a specialized application of Newton's second law.

🕒 The following is covered in Section 5.6 Quasi coordinates but omitted in our course.

Extension of the Lagrange's equation to rigid body dynamics is more complicated and will not keep the original format of Lagrange's equation.

This usually involves the introducing of **quasi coordinates** that are directly defined through velocities but their integrals usually do not a direct physical meaning as a generalized position coordinate.

This treatment is usually too confusing and not practically useful. For example, for attitude dynamics, we can directly use Euler's equations.

(Refer to textbook for more details.)

Summary

Lagrangian dynamics does not prescribe a specific set of generalized coordinates. The choice of generalized coordinates is an art, guided by physics and mathematical simplicity. Some general guidelines are:

1. Follow system symmetries to simplify potential energy.
2. Use coordinates that eliminate constraints to avoid extra equations.
3. Ensure a simple kinetic energy form to make equations manageable.
4. Choose coordinates that reveal conserved quantities for easier solutions.